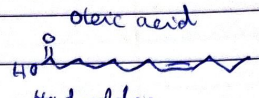
Biohybrid Nanomaterials Week 11

Bionanotubes as construction material

Motivation:

- → Protein containers as building blocks
- → NP superlattices with proteins
- → DNA as a linker for NP assembly
- → DNA as a linker for NP superlattices



Self-assembly approaches for NPs

Principle:

Hydrophobic ligands are used to stabilize and control the assembly of NP. These ligands create a hydrophobic environment that promotes the aggregation of NPs through Van der Waals interactions and hydrophobic effects.

Motivation NP superlattices

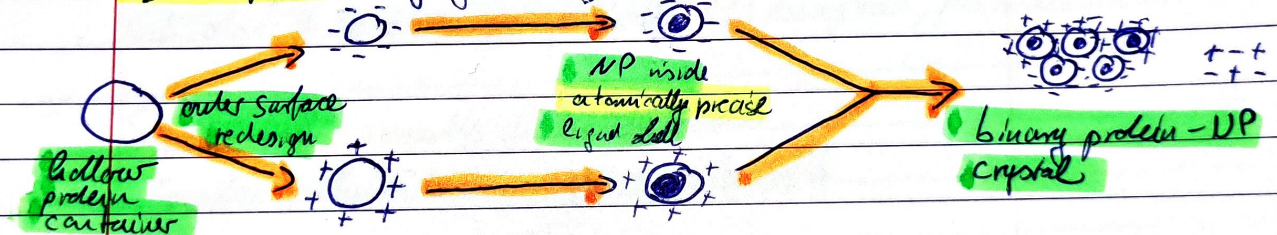
⇒ New building blocks: Protein Containers — atomically defined shell

challenges:

- → inhomogeneous particle size and morphology
- → non-precise assembly
- → small lateral dimension

Strategy for assembly of binary nanoparticle materials

I) Surface Charging — II) NP synthesis — III) Self Assembly



Biological Nanotech WiSe 11

Strategy for assembly of binary NP arrays

Step I: Designing the building blocks:

computational design with Rosetta software

↳ Surface Charge +/−

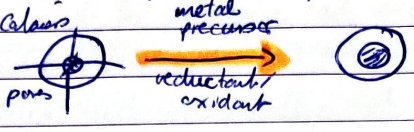
utilize natural protein containers or designed containers

+: Lys, Arg

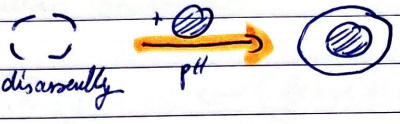
−: Asp, Glu

Step II: Loading the building blocks with NPs

charged cavity



: A



: B

Loading of charged protein containers with

- metal NP → optical, electrical, catalytic properties
- metal alloy NP → magnetic, catalytic properties
- metal oxid NP → magnetic, catalytic properties
- semiconductor NP → optical, electrical properties

but also: other cargo possible, e.g.: enzymes, pharmaceuticals

Step III: Self-assembly of building blocks

Transfer methods from protein crystallography

Ionic strength of solution			
low	optimal	high
↓ amorphous precipitate	+ − + − + − + −	↓ no assembly

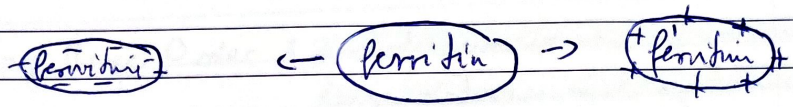
=> Advantage: High throughput! Tune parameters (salt concentrations)

Bidyfinale Handbook WP4

Results: Surface charging of ferritin with Rosetta Software

Goal → modify surface charge of ferritin proteins to facilitate their use as building blocks of nanomaterial assembly
↳ fixbb module of Rosetta

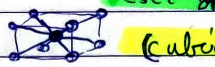
Notes:
→ unique cost for designing each residue type:
the computational design process considers the unique energetic cost associated with substituting each residue type, ensuring optimal design
→ symmetric carbonates



Crystallization
- Initial robotic screening: empty carbonates for initial screening
  high throughput: ~1500 conditions screened
- Optimization: hanging drop
- Reproducible Conditions
- Data collection for X-ray crystallography
- structure solution and refinement

Biophysical Methods Week 11

Binary Crystal Structure



transfer to tetragonal

Space group: $P4_2$

Binary protein container structure (tetragonal)

⇒ first protein crystal structure with two charged variants of the same protein

- Long range order of binary protein structure can extend the unit cell into 3D: How would you describe the arrangement of the negatively charged or positively charged container by themselves?

Long Range Order: periodic arrangements of components over large distances in a crystalline structure.

↓

OF Binary Crystal: tetragonal symmetry

$F^{+/-}$ x CO_3O_4, CeO_2, FeO_x, empty = 8 combinations

+/- with 4 different cargos ⇒ 8 combinations ⇒ Characterization with optical microscopy, SEM & EDX

Superlattice of binary protein - NP crystals: SAXS

⇒ Protein scaffold determines the structure.
NP cargo can be precisely positioned

Protein Scaffold: Thermal Stability and Modification

Fixed-Crystal: Crystals that have been chemically crosslinked to enhance their stability

Temperature Resistance: Fixed are more temperature resistant than Van-Fixed

Crystals stable in organic solvents as well

Bibliography Nanostructures with M

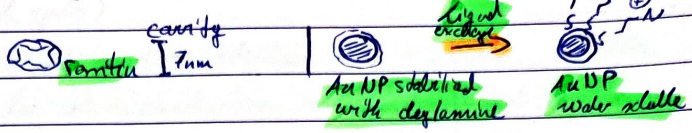
Catalysis : Cerium Oxide Ce

In Solution : Colorless ($CePt_n^-$, $CePt_n^+$)

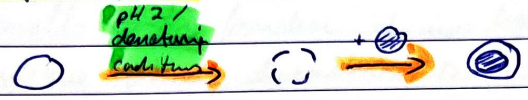
In Crystallo : Blue

Ferritin : Encapsulation of gold NPs

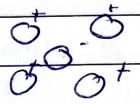
Gold NP synthesis



Encapsulation

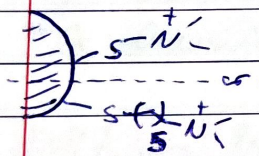


Superlattice assembly



Nanoparticle Synthesis

Control surface charge through ligand shell composition.



-> thin as possible

-> water solubility!

Why positively charged ? • Electrostatic Stability, ⌈ (strong superlattice)

• Interaction with Negatively Charged Substrate

~~Ferritin cage is negatively charged~~

Why thin ? -> Minimizing interference

↳ closer packing

-> Max active surface area

↳ enhance catalytic, optical or electronic properties

-> Reducing distance for energy transfer

Biological Nanostructures 2024 11

Plasmon – Exciton Coupling

Plasmon – Exciton Coupling: refers to the interaction between the localized surface plasmons of metallic NPs and the excitons of semiconductor molecules.

Plasmon: are collective oscillations of free electron gas density in materials (hot spots)

Exciton: are bound states of an electron and a hole, which are created in semiconductors when they absorb light

→ Results in the formation of new hybridized states that exhibit unique optical characteristics

Gold NP + Rhodamine B

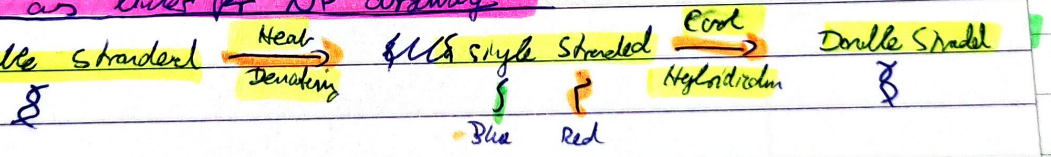
DNA as construction material

A: Adenine – T: Thymine

G: Guanine – C: Cytosine

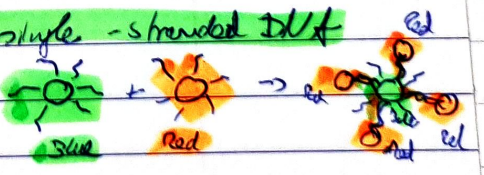
DNA as linker for NP assembly

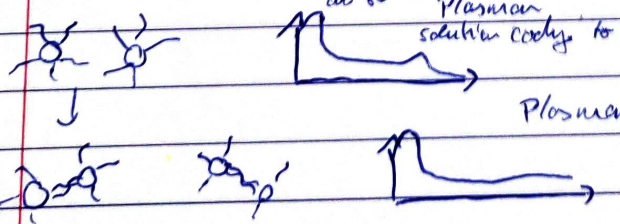
Double stranded —Heat Denaturing→ single Stranded —Cool Hybridization→ Double Stranded



→ NP can be functionalized with single-stranded DNA

→ Assembly / aggregation of NPs





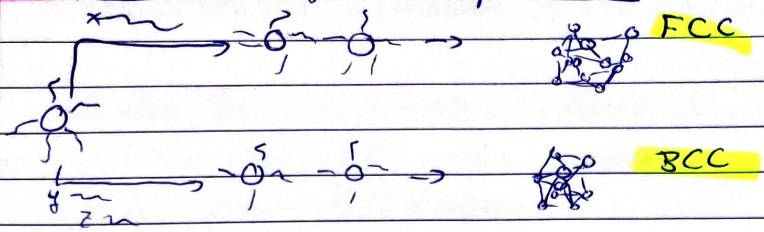
// without DNA

// With ssDNA

↳ since ssDNA hybridized! at 25°C

Biofunktionelle Materialien Woche 11

DNA as linker for NP superlattices



DNA is programmable! ⇒ control over spatial arrangement

Conclusion

→ Redesigned protein containers for assembly of NPs

→ Protein Matrix can be formed with assembly conditions

→ Highly order of NP's, fixation (stabilization) easily possible

→ DNA as a linker for NP assembly & aggregates & superlattices

→ DNA structures only stable in aqueous solution

Biohybrid Nanorobotics WS 24/25

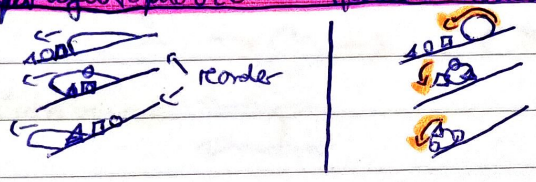
Organization of nanostructures on surfaces

- -> DNA origami
- -> Self-assembled monolayers
- -> Scanning tunnel microscopy
- -> Electron beam lithography

Motivation for nanostructures on surfaces

- -> Modification of the surface properties:
  - ↳ hydrophilic / hydrophobic
  - ↳ superhydrophilic / superhydrophobic - cleaning effects
- -> Functionalization of surfaces: -> sensors
  - -> catalysts
  - -> enhancing biocompatibility

Superhydrophobic Surfaces: Lotus Effect



Lotus Effect: phenomenon in nature where surfaces exhibit extrem water repellency

↳ superhydrophobic

↳ self-cleaning: water droplets can easily roll off the surface, picking up dirt and contaminats along the way

↳ Surface: Microscale Roughness: surface is not smooth

↳ micro/nanostructures

Application:

- -> Silicone-resin based house paint with lotus effect
- -> micro structuring through bulking agent

Bioinspired Nanostructures WdH

DNA origami

-> Scaffold: viral DNA from M13 bacteriophage
-> Staple strands self-assemble the scaffold

- **DNA origami:**
a technique used to create nanoscale structures and shapes by folding a long (single) strand DNA into specific, predetermined forms.

↳ Scaffold Strand: central component of DNA origami is a long (single) strand of DNA

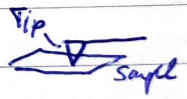
↳ Staple Strand: designed to bind to specific sequences on the scaffold strand -> folds scaffold

a) DNA Origami Formation

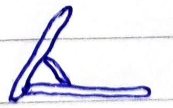
=> 3D

b) Crossover:

- **Visualization:** Atomic force microscopy (AFM)
↳ Scanning of surface with a probe tip (Si_3N_4)
↳ Detection of force between sample and tip
↳ Electronics control constant height
↳ Signal: Force between tip and sample

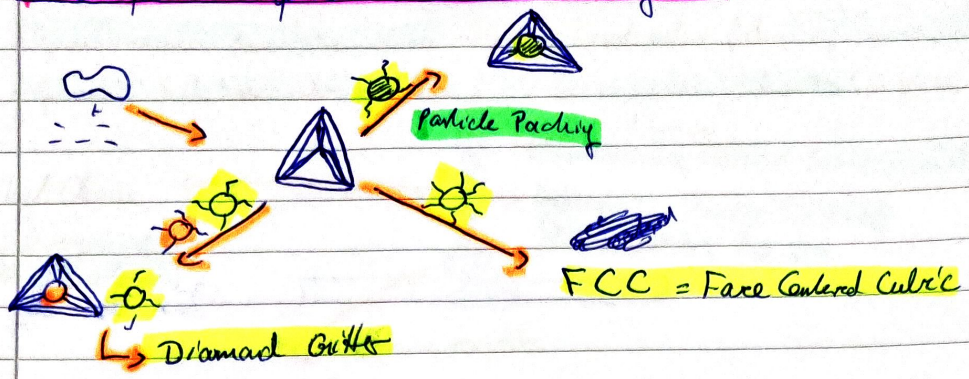


$F = k \Delta z$
k = spring constant
Δz = deflection of the cantilever

- can form: ☆ ☺ ♡
- DNA nanotechnology 

Bibliography Nanostructure with DNA

Nanoparticles superlattices with DNA origami



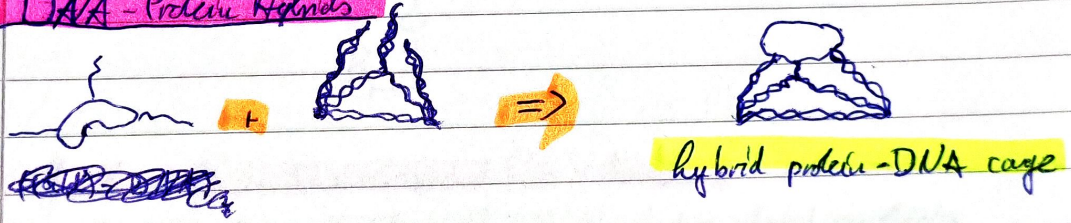
Why two type of Gold NP?

→ highly specific assembly → create diamond structure

→ plasmonic coupling

→ stabilization

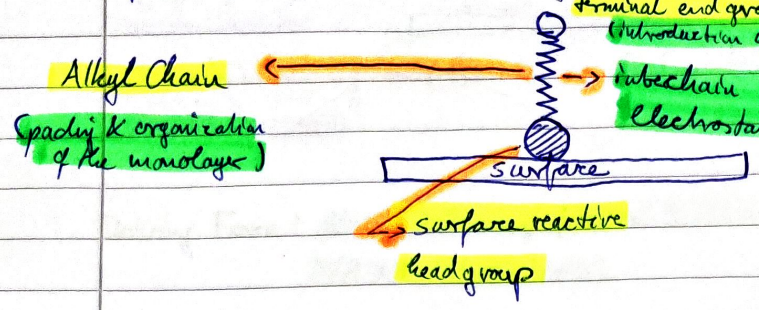
DNA - Protein Hybrids



Biofluide Nanostructure 2014 12

Self-Assembled Monolayers (SAMs)

Spontaneous formation of a mono-molecular film by immersion of solid substrate into solution of surface-active molecules (surfactant)

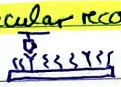


Applications :

→ wetting :

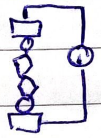
→ biosensitive systems

→ Molecular recognition :



→ Micro & Nano-fabrication :

→ Nano-devices & Molecular electronics :

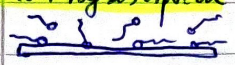
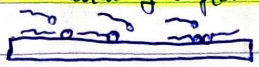
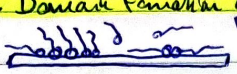


Self-Assembled Monolayers : Surfactants

- → Alkoxysilanes / Chlorosilanes on hydroxylated surfaces
- → Alkyl thiols on Gold
- → Alcohols & Amines on Gold, Pt
- → Carboxylic Acids on Al_2O_3

Biological Monolayers

Assembly process and SAM formation

1. Physisorption

2. Formation of a flat lying phase

3. Domain Formation of Standing Phase

4. Formation of a densely packed standing phase

Driving Force: Minimization of free surface energy of the substrates, VdW interactions.

Parameters:

intrinsic ⇒ CH_2 chain length, crystallographic orientation of substrates, defects

extrinsic ⇒ temperature, concentration, time, solvent

↑ increase self assembly, but also defects

↳ denser

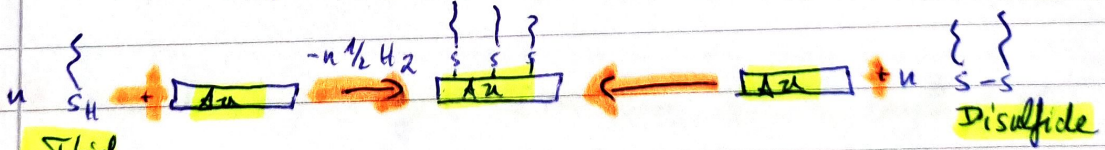
thiol = $R-S-H$ disulfid = $R-S_1-S-R$

Self-Assembled Monolayers of thiols / disulfides on gold

First Studies: Chemisorption of thiols / disulfides by oxidative addition of SH-bond and reductive elimination of hydrogen (thiolate formation) $CH_3(CH_2)_n S-Au + 1/2\ H_2\uparrow$

Binding strength: Au-S : ≈ -200 kJ/mol

↳ Formation of SAM from liquid phase or gas phase



Thiol

Why are thiols forming more highly ordered SAMs compared with organosilanes?

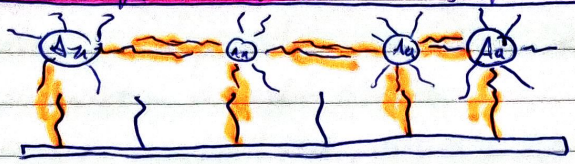
↳ Stronger and more specific binding, than organosilanes on oxides

↳ Denser packing, organosilanes are tend to form less densely packed

↳ smaller ↳ direct chemisorption, org. need multiple steps

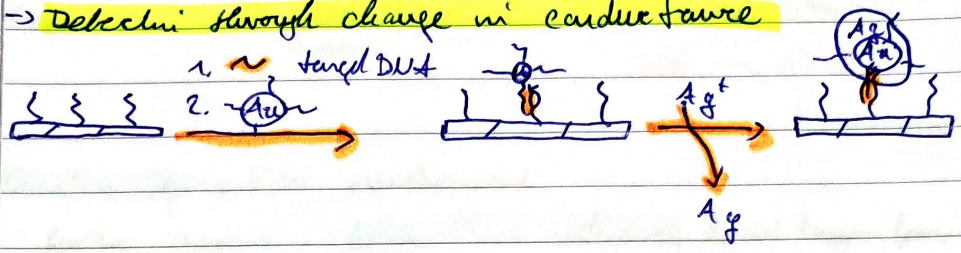
Biohybride Nanostructures WS 12

DNA as linker for NP attachment at surfaces



DNA assay detection with NP

-> Detection of DNA
-> Gold (Au) NP bind to surface via target DNA
-> Silver (Ag) ions are reduced to silver NPs
-> Detection through change in conductance

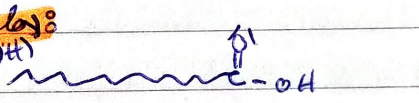


Langmuir Blodgett (LB)

-> Assembly of amphiphilic molecules at air-water-interface
-> Works with particles as well: e.g. SiO_2 particle layer, photonic crystals
-> Stable monolayer through mechanical pressure

eg (Amphiphilic molecules):

-> Fatty acids ($RCOOH$)
-> Amines (RNH_2)
-> Phospholipids



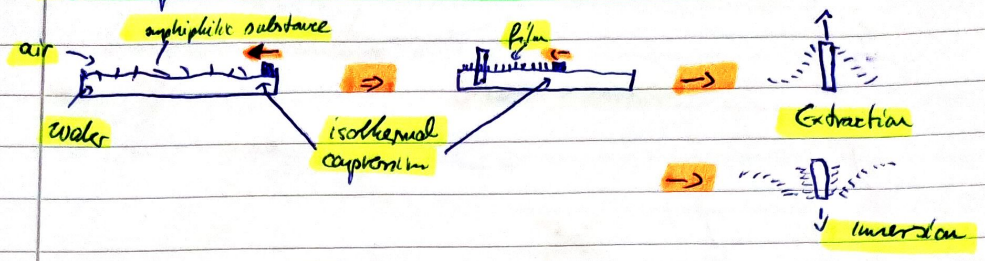
=> High control over layer thicknesses and distances

=> Parameters: polarity of the end group, pH, T, length of the unpolar groups ($n > 11$)

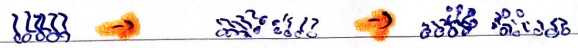
Biological Nanostructures 2014/2

Langmuir Blodgett

-> Dissolve the amphiphilic component in volatile solvent (eg. chloroform)
-> Spread this solution on water surface
-> After evaporation of solvent, surface is reduced with barrier blocks: Compression of monolayer
-> Transfer to substrate via immersion or extraction

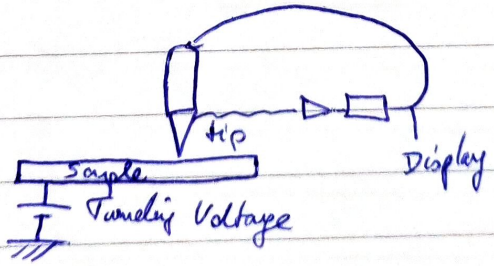


- Creation of synthetic membranes:
  -> further compression & monolayer collapses, double layer forms



Scanning tunneling Microscopy (STM)

-> No - touch scanning of surface with fine metal tip (Pt/Ir, W)
-> Potential applied or tunnel current is measured
-> Constant height mode: height is kept constant but tunnel current changes
-> Constant current mode: changes in control voltage are mapped
-> No direct mapping of individual atoms convolution of electronic states with topography



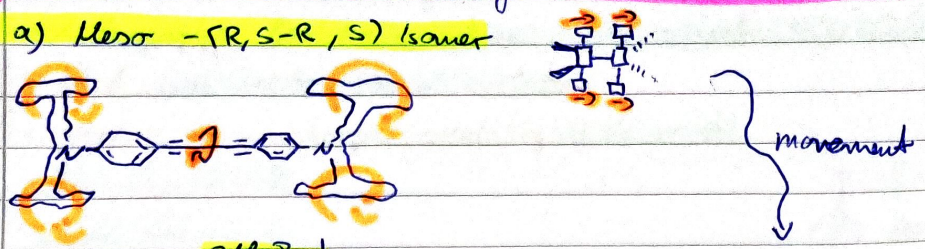
SAM:

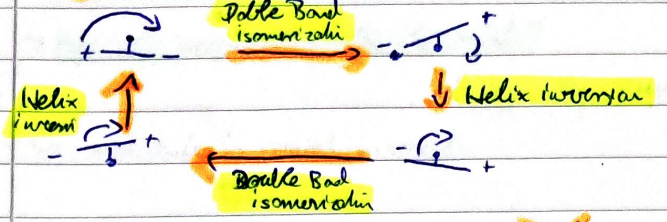
Low concentration:	High concentration:
Molecules lying flat on the surface, striped phase	Molecules are standing up, formation of dense monolayer

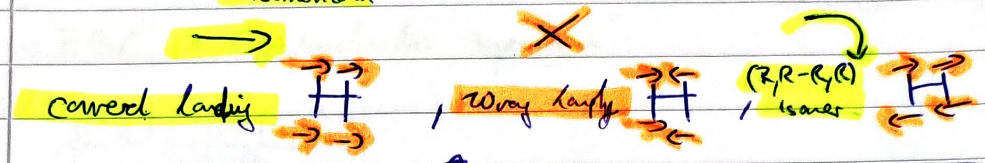
Bistyfid Nanotubes

Molecular Machines & Electrically driven directional motion

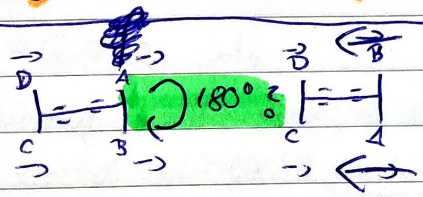
a) Meso - (R,S-R,S) isomer







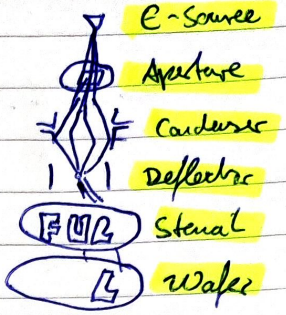
What if



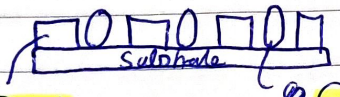
Biohybrid Nanosystems WiHi 12

Electron Beam Lithography (EBL)

-> Lithography for the creation of nanostructures on surfaces
-> EBL has sub-10 nm resolution
-> electron beam changes solubility of the resist
-> low throughput
-> high vacuum
-> special requirements for the resist
↳ typically PMMA
↳ substrate is coated in resist



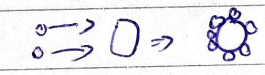
-> EBL + Electrophoretic Deposition



PMMA ◎ Gold Nanorods

-> Gold Nanorods array scatters light in angle dependent manner
-> Gold Nanorods via electric field: electrophoretic deposition
↳ Higher resolution

↳ used to deposit charged particles from a colloidal suspension onto a substrate under the influence of an electric field

eg: 

Other Lithography Techniques:

-> soft lithography
-> Nano imprint lithography
-> Nano sphere lithography
-> ...

Bachelor Handout 2024/25

Conclusion

-> STM & AFM for characterization of surfaces, but also for manipulation / structuring

Biohybrid Nanomedicine Week 13

Application for NP and biohybrid nanomedicine

Motivation

→ Nanoscale materials have similar dimensions as cells: facilitate interactions

→ Nanoparticles have a large surface / large cargo capacity: enhance drug loading

→ Treatment and Diagnostic: Theranostics

Protein-based systems:

→ defined particle size and defined ligand shell result in defined pharmacokinetics

→ functionalization and engineering

→ Biocompatible

→ Biodegradable

eg: Magnetic iron oxide inside ferritin → cancer recognition peptide on the outside

Ferritin for drug delivery

→ Human heavy chain ferritin with RGD peptide attached to N-terminus (RFRT)

→ loaded with doxorubicin: complexation with Cu, incorporation by diffusion

⇒ reduces growth rate of cancer cell

Ferritin for photodynamic therapy

→ RFRT

→ loaded with photosensitizer zinc hexadecafluorophthalocyanine

⇒ reduces growth rate of cancer cell

Biohybrid Nanostructures WS 24/25

Photothermal imaging with Ferritin - Cu_2S NP

→ Synthesis of Cu_2S NPs inside Ferritin

→ Temperature increase after laser irradiation & at different concentrations

→ bind to tumor ⇒ photothermal imaging → locate tumor

Problem - Polymer systems

- Classical Drugs:
  - → low bioavailability
  - → side effects

⇒ Nanoparticles for chemotherapy

- Challenges:
  - → Degradation by proteases
  - → Liver uptake at clearance

⇒ Integration of ferritin into microgels

- Why ferritin for drug delivery?
  - → Biocompatible & robust
  - → Open/close - able structure
  - → Surface modification
  - → Cancer targeting properties
- Advantages of microgels?
  - → High uptake capacity
  - → Easy - controllable synthesis
  - → Multifunctionality
  - → Stimuli - responsive systems

Synthesis of Ce-Ftn microgels:

What is the advantage of adding Ferritin during the microgel synthesis vs adding it to already synthesized microgels?

→ even form distribution

↳ increased drug loading efficiency

⇒ Integration Strategy: Electrostatic interactions

Biophysical Chemistry WiSe 23

Encapsulation of fluorophores and doxorubicin

→ colors

→ different pH can lead to release of cargo of ferritin
↳ add acid

→ Proving that microgels protects ferritin:
1) Digestion of ferritin-microgels by chymotrypsin
3) Check ferritin release in pH 2.5

⇒ Microgel protects ferritin against degradation by protease!

Gold NP

→ Size-dependent toxicity of Gold NPs!

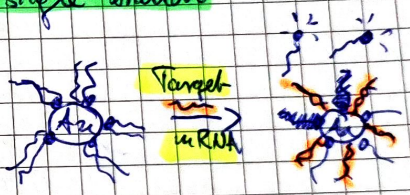
DNA nanoflares

→ Gold nanoparticle decorated with DNA single strands (spherical nucleic acids, SNA)

→ Recognition sequence bound to NP

→ Reporter flare is bound but quenched

→ Reporter flare released upon binding of target mRNA; shows fluorescence



Reporter Flare
≈
Recognition Strand

Does the reporter flare need to bound more strongly or more reversibly to the recognition strand compared with the mRNA?

→ more reversibly ⇒ easily displace by mRNA

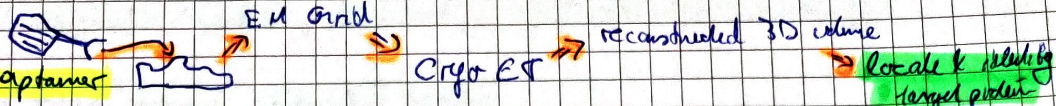
⇒ for detection of cancer cells in blood!

Biological Nanowires

DNA origami to recognize proteins

→ DNA origami + RNA aptamer binds to protein target

→ Proof of principle: Fluorescent proteins as target



DNA origami for siRNA delivery

→ Oligonucleotide NPs (ONPs)

→ siRNA silencing: Binds to RNA in cell and silences protein expression

→ Folate molecules (bullets) are needed for cellular uptake

→ In vivo: longer blood circulation time than the parent siRNA

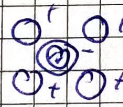
Covid Vaccines

Ferritin as central core.

mRNA encapsulated into lipidic nanoparticle

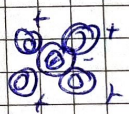
Binary NP

enzyme/Fe



no oxidase-like activity!

Co/Fe



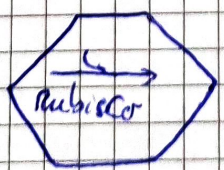
Oxidase-like activity!

Natural compartments for Catalysis

Bacterial microcompartment:

→ Icosahedral, polyhedral carboxysome

→ encapsulates enzymes: higher efficiency, specificity



→ optimization of reaction conditions

Why Rubisco in Microcompartment?

↳ concentration of substrates → regulation & control

Biophysik Vorlesung 28.11.19

Conclusion

→ Protein-based systems have several advantages for biomedical applications & combination of biomolecule with functional molecule / cargo, combination of diagnostics and therapy

→ DNA origami: structure design

→ Challenges for biomedical applications are stability, clearance, toxicity, biodistribution

→ For material applications: issue of final yield and cost of the materials